

SUMMER COOLING LOAD ANALYSIS & HVAC DESIGN

THE GRAD HOUSE - HALPERN GSC BUILDING #160

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PROJECT BACKGROUND

Renovation of the Halpern Centre for Graduate Students (Bldg #160) into an open-concept student social space: restaurant/lounge, bar, meeting rooms, and office. A new rooftop constant-volume AHU serves all 15 conditioned spaces via a single supply and return duct system.

Three Architectural Modifications:

- Removed vestibule–lounge partition wall, opening entry directly into Rm 110
- Reconfigured Storage 108A to connect to Meeting Rm 108 (not hallway)
- Extended Storage Rm 109 into corridor footprint

DESIGN CONDITIONS

Parameter	Value	Source
Outdoor DB (1% annual)	26.6 °C	ASHRAE 2021 [2]
Outdoor WB	19.4 °C	ASHRAE 2021 [2]
Indoor Setpoint	24 °C / 50% RH	ASHRAE 55 [10]
Supply Air Temp	13.0 °C	11°C delta-T
Wall U-value	0.28 W/m²K	90.1-2019 [4]
Roof U-value	0.19 W/m²K	90.1-2019 [4]
Window U / SHGC	2.56 / 0.40	90.1-2019 [4]
Sol-Air ΔT - Roof	31.0 °C	ASHRAE Ch.18 [1]
Sol-Air ΔT - W/E	30.4 °C	ASHRAE Ch.18 [1]

COOLING LOAD SUMMARY

Heat Gain Component	Sensible (W)	Total (W)
Solar - West & SE glazing	11,160	11,160
Lighting (LED, BF=1.00)	8,164	8,164
Occupancy - Sensible	5,225	5,225
Envelope - Roof	2,750	2,750
Envelope - Walls	2,065	2,065
Equipment - Sensible	1,892	1,892
Occupancy - Latent	-	3,615
Other (infiltr., windows, equip. lat.)	785	785
Subtotal (pre-safety factor)	31,717	35,916
DESIGN TOTAL (×1.10 SF)	34,889	39,508 W

VENTILATION

$$V_{bz} = R_p \times P_z + R_a \times A_z$$

- Office/conference: $R_p = 5.0 \text{ L/s} \cdot p$, $R_a = 0.06 \text{ L/s} \cdot m^2$
- Lounge/restaurant/bar: $R_p = 3.8 \text{ L/s} \cdot p$, $R_a = 0.06 \text{ L/s} \cdot m^2$

Min. Outdoor Air: 554 L/s = 27.7% OA fraction

HVAC SYSTEM DESIGN

$$Q_{\text{supply}} = Q_{\text{s,design}} / (\rho \cdot c_p \cdot \Delta T) = 34,889 / (1.20 \times 1006 \times 11.0)$$

Design Supply Airflow: 2,002 L/s at 13°C

AHU cooling coil: 47.3 kW (13.5 tons). Heating from UVic district energy hot water loop.

DUCT DESIGN

- Target friction rate: 0.8 Pa/m Throughout (ASHRAE Fund. Ch.32 [1])
- Darcy $f = 0.02$ - galvanised steel, smooth interior
- Main trunk $\leq 7.5 \text{ m/s}$ · Branch $\leq 6.0 \text{ m/s}$ · Final run $\leq 5.0 \text{ m/s}$
- Aspect ratio $\leq 4:1$ · Sizes rounded to nearest 25 mm

Description	Flow (L/s)	W×H (mm)	Vel (m/s)
Main trunk (AHU→Rm112)	2,002	575×575	6.05
After Rm 112 takeoff	1,637	525×525	5.94
After Rm 111 takeoff	1,446	500×500	5.78
Branch → Rm 110 Restaurant	899	425×400	5.29
Left branch → Hallway 103	386	300×300	4.29
Final run → Rm 106A	80	175×150	3.06
Final run → Rm 108A (critical)	8	125×100	0.64

DIFFUSER SELECTION

$$L = (\text{half-zone side}) + (h_{\text{ceil}} - 2.74 \text{ m}) \quad | \quad T_{0.25/L}: 0.5\text{--}1.5 \quad \checkmark$$

FAN SELECTION

Path	Route	Duct ΔP	Vel. Regain	Diff. SP	Total
1→7	AHU → Rm 101	66.3 Pa	−1.7	10 Pa	74.7 Pa
1→10	AHU → Rm 106A	64.2 Pa	−2.4	10 Pa	71.8 Pa
1→13	AHU → Rm 108A	68.7 Pa	−4.1	23 Pa	88 Pa ✓

2,002 L/s
Design Flowrate

88 Pa
External Static Pressure

MANUAL VS. EQUEST

Parameter	Values	Δ
Peak Cooling Load	39.5 kW manual	-
Annual Cooling Energy	19,632 vs. 19,650 kWh	+0.1%
Annual Lighting Energy	55,770 vs 56,237	+0.8%
Equivalent Full Load Hrs	497 h/yr (derived)	-

OPERATING COSTS

\$11,250

Electricity/yr
160,721 kWh × \$0.07

\$15,434

Natural Gas/yr
1,110 GJ × \$13.9

\$26,684/yr total

Heating = 57.9% of bill. Cooling = only 5.2% - reflecting Victoria's mild climate.

CARON NEUTRAL & CONTROLS STRATEGY

- District energy decarbonization:** UVic's transition to low-carbon heat source eliminates building-level heating carbon footprint
- ERV on AHU OA intake:** 70% effectiveness recovers ~620 MBTU/yr heating → saves ~\$9,200/yr
- 300 m² rooftop PV array:** ~54,000 kWh/yr generated → saves ~\$3,780/yr
- Occupancy setbacks:** Cooling to 27°C / heating to 18°C during ~5,000+ unoccupied hours/yr
- Demand-controlled ventilation (DCV):** CO₂ sensors in Rm 110 + 112 modulate OA damper per ASHRAE 62.1-2019 [5]
- SAT reset:** Supply air reset 13→16°C at part-load to reduce compressor and fan energy
- Lighting occupancy sensors:** Meeting rooms + office; lighting = 35% of electricity use

ERV + PV combined savings: ~\$13,000/yr → total cost ≈ \$13,700/yr (49% reduction)